BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा

स्थायी विरंजक पाउडर — सुरक्षा सहिंता

(*IS 15738 का पहला परीक्षण*)

Draft Indian Standard

Stable Bleaching Powder — Code of Safety

(*IS 15738 of First Revision*)

ICS 71.060.50

Chemical Hazards Sectional Committee, CHD 7

Last date for Comments: As per the covering letter

FOREWORD

(*Formal Clause shall be added later*)

Stable bleaching powder (SBP) is a white or nearly white powder, that is, usually a mixture of calcium hypochlorite, calcium chloride and calcium hydroxide. It is also called chloride of calcium hypo and is prepared by reacting calcium hydroxide or slaked lime with chlorine gas. It is used as a strong bleaching agent for paper and textiles, as a disinfectant, algaecide, bactericide, deodorant, potable water purification, disinfectant for swimming pools, etc.

The elimination of accidents is vital to public interest. Accidents produce economic and social loss and impair individual or group productivity. Realization of this loss has led the authorities to devote a good deal of attention to safety education. In any programmed of safety education, preparation of code of safety is an essential part. Apart from general precautions, some typical precautions are required to be taken and this code of safety lays special emphasis on these points. For, proper utilization of the code of safety for stable bleaching powder, a knowledge of effects of hazardous substances on biological systems is desirable. This code of safety recommends practices to be followed to ensure safety of personnel engaged in industries where carbon tetrachloride is produced, stored, handled or used.

This standard was originally published in 2007. The Committee felt a need to revise this standard with a view to update the standard based on the experience of last one decade and on the currently available data.

In this revision general properties have been incorporated, and modifications have been made to update safety measures for controlling hazards and essential information on symptoms of poisoning, first-aid, medical treatment, storage, handling, labelling and employee safety, fire and explosion hazards, routes of entry, toxicity information, fire prevention and firefighting.

There is no standard on this subject published by ISO. This standard has been prepared indigenously based on the available data and information.

The various clauses of the standard have been aligned with the format being applied for all Indian Standards on code of safety of chemicals.

The properties of stable bleaching powder listed in Clause **4** have been taken from literature and have been included for information only. Moreover, these properties pertain to carbon tetrachloride. BIS has published a separate standards IS 1065 (Part 1) : 2019 on the requirements and the methods of sampling and test for stable bleaching powder intended for household and industrial use on the requirements and the methods of sampling and test for carbon tetrachloride intended for industrial purposes and IS 1065 (Part 2) : 2019 on requirements and the methods of sampling and test for stable bleaching powder for treatment of water intended for drinking.

1 SCOPE

1.1 This standard provides properties of stable bleaching powder (SBP) and the nature of hazards associated with it.

1.2 This standard also covers other essential information such as information on storage, handling, packing, labelling, disposal of waste, cleaning and repair of containers, selection and training of personnel, protective equipment and first-aid.

1.3 This standard does not deal with specification for design of buildings, chemical engineering plants, storage vessels, equipment for operations control and waste disposal.

2 REFERENCES

The standards listed in **Annex A** contain provisions which through reference in this text, constitute provisions of this standard. At the time of publications, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

3 TERMINOLOGY

For the purpose of this standard, the definitions given in IS 4155 shall apply.

4 PROPERTIES

4.1 General Information

4.1.1 *Chemical Name* **—** Calcium hypochlorite.

4.1.2 *Common Name and Synonyms* **—** Stable bleaching powder.

4.1.3 *Uses*

Stable bleaching powder (SBP) is widely used as a disinfectant, bleaching agent, and oxidizing agent due to its high available chlorine content and excellent storage stability. It is extensively used for the disinfection of drinking water, swimming pools, wastewater, and sewage to eliminate harmful microorganisms and ensure public health. In healthcare facilities and households, it is used for sanitizing floors, toilets, drains, equipment, and other surfaces to control the spread of infectious diseases. In the textile and paper industries, stable bleaching powder serves as an effective bleaching agent for fabrics and wood pulp, while in food processing it is used for sanitizing equipment and processing areas. It is also applied in aquaculture and agriculture for disinfecting ponds, irrigation water, and equipment, as well as in emergency and disaster relief operations for providing safe drinking water. Additionally, stable bleaching powder is used in industrial cleaning and various chemical processes as an oxidizing agent and for controlling odours in sewage and waste disposal systems.

4.2 Identification

4.2.1 *Formula* **—** $CaOC1_2$.

4.2.2 *CAS Number* **—** 7778-54-3.

4.2.3 *UN Number* **—** 2208 (more than 10 % but not more than 39 % available chlorine).

4.2.4 *UN Class* **—** 5.1 (oxidizing substance).

4.3 Physical Properties

4.3.1 *General* **—** White powder.

4.3.2 *Molecular Mass* **—** 142.98 g/mol.

4.3.3 *Physical State* **—** White/grey solid at room temperature.

4.3.4 *Colour* **—** White/grey.

4.3.5 *Odour* **—** Chlorine like odour.

4.3.6 *Boiling Point* **—** at 760 mm Hg, decomposes at 100 ºC.

4.3.7 *Melting Point/Freezing point* **—** Not applicable.

4.3.8 *Relative Vapour Density* (*Air*) – Not applicable.

4.3.9 *Specific Gravity* **—** 2.35 at 20 ℃.

4.3.10 *Viscosity at 30 °C* – Not applicable.

4.3.11 *Vapour Pressure* **—** Not applicable.

4.3.12 *Heat of Combustion* (Higher Value) — Not applicable.

4.3.13 *Refractive Index* (at 0 ℃ and 1 atm) — Not applicable.

4.3.14 *Solubility in Water* **—** 21.4 percent at 25 °C and 23.4 percent water at 40 °C.

4.3.15 *Solubility in other solvents* — Not soluble.

4.3.16 *Light Sensitivity*

Stable bleaching powder is light-sensitive and undergoes gradual decomposition when exposed to direct sunlight or ultraviolet radiation. Light accelerates the degradation of calcium hypochlorite, resulting in a progressive loss of available chlorine, which reduces the product's bleaching and disinfecting effectiveness. Prolonged exposure to light, especially when combined with heat and moisture, further increases the rate of decomposition and may lead to the release of chlorine-containing gases. Therefore, stable bleaching powder should be stored in tightly sealed, light-resistant containers in a cool, dry location to maintain its stability and shelf life.

4.3.17 *Decomposition Temperature*

Stable bleaching powder undergoes thermal decomposition when heated to approximately 175 °C to 180 °C. The decomposition is exothermic and results in the release of oxygen and chlorine-containing gases, reducing the available chlorine content and diminishing its disinfecting and bleaching effectiveness. The rate of decomposition increases significantly in the presence of moisture, direct sunlight, acids, reducing agents, or organic contaminants.

4.4 Chemical Properties

4.4.1 *Reactivity*

4.4.1.1 Hypochlorites yield hypochlorous acid (HOCl) when dissolved in water and are converted to Cl_2 when mixed with acid. They react quantitatively in acid media with iodide liberating iodine and with hydrogen peroxide (H_2O_2) liberating oxygen.

4.4.1.2 The stability of SBP is a function of its moisture, time and impurity content as well as the temperature and humidity at which it is stored. It rapidly decomposes on exposure to air and may decompose violently, if exposed to heat or direct sunlight. It is thermally unstable and decomposes above 170 °C giving different reaction products. SBP reacts vigorously with nitromethane, reducing materials and react with sulphur and violently with organic matter above 100 °C.

4.4.2 *Polymerisation* **—** Not applicable.

4.4.3 *Allotrope Formation* **—** Not applicable.

4.4.4 *Corrosion Properties* **—** Corrosive to most metals in the presence of moisture.

4.4.5 *Critical Constants* **—** Not available.

4.4.6 *Hazardous Decomposition Products*

SBP reacts with water or steam to produce toxic and corrosive fumes of hypochlorous acid. Above 170 °C, it undergoes exothermic decomposition forming calcium chloride and releasing oxygen. Traces of water may initiate the reaction. A rapid exothermic decomposition above 175 °C releases oxygen and chlorine.

4.4.7 *Incompatibilities*

Calcium hypochlorite is a strong oxidizer. It reacts with water and acids giving off chlorine gas, forms explosive compounds with ammonia and amines. Incompatible with organic materials, nitrogen compounds and combustible materials. In the presence of moisture, it is corrosive to most of the metals.

4.5 Fire and Explosion Hazard Properties

4.5.1 *Ignition Temperature* **—** No data available.

4.5.2 *Auto Ignition Temperature* **—** No data available.

4.5.3 *Flash Point* **—** No data available.

4.5.4 *Upper Explosive Limit* **—** No data available.

4.5.5 *Lower Explosive Limit* **—** No data available.

4.5.6 *Fire Risk*

4.5.6.1 It is a powerful oxidizing agent. Not combustible but supports combustion of other materials. Decomposes violently upon heating liberating oxygen, hence supports any fire. It reacts with organic materials can readily ignite combustible materials. Decompositions can be rapid and violent upon contact with incompatible materials and on heating.

4.5.6.2 Sealed containers of SBP may rupture when heated. An explosion may occur if either a carbon tetrachloride or a dry ammonium compound fire extinguisher is used to extinguish a fire.

4.5.7 *The Limits of Detonability in Air*

4.5.8 *Flammability*

Stable bleaching powder is non-combustible; however, it is classified as a strong oxidizing substance. Although it does not burn, it can support and intensify the combustion of combustible materials by supplying oxygen during decomposition. Contact with organic materials, reducing agents, oils, greases, paper, wood, textiles, or other combustible substances may lead to vigorous reactions and increase the risk of fire. When exposed to heat or fire, stable bleaching powder decomposes, releasing oxygen and toxic chlorine-containing gases, which can further intensify combustion and pose significant health hazards.

5 HEALTH HAZARD AND TOXICITY INFORMATION

5.1 General Information

5.2 Routes of Entry

5.2.1 *Skin*

SBP is not appreciably irritating to skin in the dry form. Repeated or prolonged skin contact or contact with moist skin can result in moderate to severe burns and dermatitis.

5.2.2 *Eyes*

Contact of SBP with eyes cause eye irritation, blurred vision, redness, pain and severe tissue burns. Contact with eyes cause eye irritation, blurred vision, redness, pain and severe tissue burns.

5.2.3 *Ingestion*

Swallowing of SBP may result in gastrointestinal corrosion causing severe pain, nausea, vomiting and diarrhoea, inflammation of mouth and oesophagus. Large doses may be fatal.

5.2.4 *Inhalation*

Dusts of chlorine (decomposition product) are corrosive to the respiratory tract. It is destructive to tissues of mucous membrane and upper respiratory tract. Symptoms may include burning sensation, coughing, wheezing, laryngitis, shortness of breath, headache, nausea and vomiting. Inhalation may be fatal as a result of spasm inflammation and edema of the larynx and bronchi, chemical pneumonitis and pulmonary edema.

5.3 Toxicity information

5.3.1 The toxicity information related to stable bleaching powder is as follows:

a) Threshold Limit Value (TLV), Time Weighted Average (TWA) — 1 ppm peak limitation;

b) Short Term Exposure Limit (STEL) — 2 mg/m^3 for 15 min;

c) Lethal Dose (LD_{50}), (Rat), Oral — 850 mg/kg;

d) Odour Threshold — 0.5 ppm to 2 ppm; and

d) National Fire Protection Association (NFPA) Hazard Signal — Health (3)

5.3.2 *Peak Limitation*

A ceiling concentration that should not be exceeded over a measurement period which should be as short as possible but not exceeding 15 min. These exposure standards are guides to be used in the control of occupational health hazards. Exposure standards should not be used as fine dividing lines between safe and dangerous concentrations of chemicals. They are not a measure of relative toxicity.

5.4 Antidote

In case of stable bleaching powder poisoning, medical attention may be sought.

5.5 Health Effects

5.5.1 *Signs and Symptoms*

Inhalation, ingestion or contact (skin, eyes) with vapors or substance may cause severe injury, burns or death. Fire may produce irritating, corrosive and/or toxic gases. Runoff from fire control or dilution water may cause pollution.

5.5.2 *Acute Toxicity*

5.5.2.1 SBP has corrosive properties due to release of toxic gases such as chlorine. The reaction of hypochlorite with fats and proteins of body results in deep tissue destruction known as liquefaction necrosis. It also causes thrombosis of blood vessels. These symptoms may be observed immediately or within a few hours. While the effect of toxicity increases with hypochlorite concentration and *p*H.

5.5.2.2 Exposure to SBP through ingestion leads to pharyngeal pain, and gastrointestinal irritation including injury to mouth, throat oesophagus and stomach with bleeding, scarring, perforation. In addition, dysphagia, stridor, drooling, odynophagia and vomiting, pain in the abdomen or chest are likely symptoms. It may also result in metabolic acidosis after ingestion of household bleach. It also causes severe eye injuries with necrosis and chemosis of cornea, clouding of cornea, iritis, cataract formation of severe retinitis.

5.5.2.3 *Systemic effects*

Systemic effects occur after absorption of the chemical into the body and distribution. It exhibits relatively limited systemic toxicity. Higher concentration inhalation can lead to pulmonary edema a condition where there is accumulation of fluid in the lungs due to respiratory distress (airway constriction). The symptoms include rapid breathing, cyanosis, wheezing, rales, or hemoptysis. It can also lead to reactive airways dysfunction syndrome (RADS) in a latent period of 5 min to 15 h. Systemic toxicity is generally secondary to the corrosive effects rather than direct toxic action on specific organs.

5.5.2.4 *Local effects*

Local effects occur at the site of first contact with the chemical and are the predominant toxic effects associated with stable bleaching powder. Contact with the powder or concentrated solutions causes severe eye irritation, pain, redness, excessive tearing, conjunctival inflammation, and may result in corneal burns or permanent eye damage if not promptly treated. Direct contact of SBP with skin causes irritation, redness, burning sensation, and chemical burns. Prolonged exposure may produce blistering and tissue damage. Inhalation of dust or chlorine gas released from decomposition causes irritation of the nose, throat, and upper respiratory tract, resulting in coughing, sore throat, chest tightness, wheezing, and difficulty breathing. High concentrations may cause pulmonary edema, a potentially life-threatening condition. Ingestion of SBP causes corrosive injury to the mouth, throat, esophagus, and stomach, leading to severe pain, nausea, vomiting, abdominal discomfort, and, in severe cases, gastrointestinal bleeding or perforation.

5.5.3 *Chronic Toxicity*

Dermal irritation is caused due to prolonged dermal exposure. It may cause bronchitis that results in shortness of breath with cough. While there is no information for cause of carcinogenic, mutagenic, teratogenic effect and reproductive and developmental effects.

5.5.3.1 *Systemic effects*

Oesophageal obstruction, pyloric stenosis, squamous cell carcinoma of the oesophagus, and vocal cord paralysis with consequent airway obstruction are a few systemic effects of injury when exposed through to ingestion of hypochlorite.

5.5.3.2 *Local effects* — Not available.

6 PERSONAL PROTECTIVE EQUIPMENT

6.1 Availability and Use

6.1.1 While personal protective equipment is not an adequate substitute for safe working conditions, adequate ventilation and intelligent conduct on the part of employees working with stable bleaching powder, it is in many instances, the only practical means of protecting the worker, particularly in emergency situations. Evacuate personnel to safe areas and remain cautious of vapours accumulating to potentially form explosive concentrations, particularly in low-lying areas. Personal protective equipment protects only the worker wearing it, and other unprotected workers in the area may be exposed to danger.

6.1.2 Employees should be instructed to have minimum contact with stable bleaching powder. In any case, continuous exposure to concentration of these hydrocarbons above their threshold limit value should be avoided. Where unavoidable, proper respirators should be used.

6.1.3 The correct usage of personal protective equipment requires the education of the worker in proper employment of the equipment available to personnel.

6.1.4 Under conditions which are sufficiently hazardous to require personal protective equipment, its use should be supervised and the type of protective equipment selected should be capable of control over any potential hazard.

6.2 Non-Respiratory Equipment

Personal protective equipment should include non-respiratory equipment like chemical safety goggles or a full-face shield where splashing is possible. Dust masks should be used, while handling the powder to prevent its entry in breathing. Overall, safety shoes, gloves, apron or as appropriate should be worn by employees working in the area to avoid skin contact.

6.2.1 *Eye and face Protection*

Chemical splash protection as per IS 8520 category H-4. Wear safety goggles, self-contained breathing apparatus, airline mask, hand gloves.

6.2.2 *Head Protection*

Hard hats should be worn where there is danger of falling objects. Safety helmets (see IS 2925) should be worn where there is danger from falling objects.

6.2.3 *Foot and leg Protection*

Leather safety shoes and leather leg guard is recommended. Shoes should be thoroughly cleaned and ventilated after contamination (*see* IS 10667).

6.2.4 *Body, Skin and Hand Protection*

Leather or rubber gloves should be worn for hand protection. Fireproof overalls should be worn when operations involving fire hazards are encountered (*see* IS 8519 and IS 8807).

6.3 Respiratory Equipment

As there is a risk of exposure to chlorine, suitable respiratory protective devices should be used. For emergencies or instances where the exposure levels are not known, a self-contained breathing apparatus operated in positive pressure mode, must be used.

7 STORAGE, HANDLING, LABELLING AND TRANSPORTATION

7.1 Storage Unloading Trucks/Drums

7.1.1 *General Information*

7.1.1.1 SBP should be stored in a cool, dry and well-ventilated area. In areas where natural ventilation is not possible, forced ventilation equipment should also be installed.

7.1.1.2 SBP can react vigorously and sometimes explosively with certain organic and inorganic materials (*see* **4.3**), therefore all incompatible materials should be kept away from storage area.

7.1.1.3 SBP may ignite or explode on storage. Traces of water may initiate reaction. It should be stored properly to avoid contact with heat, moisture, flames and sparks.

7.1.1.4 This material is a scheduled poison. Foodstuffs should be stored away from it.

7.1.1.5 Material should not be stored on wooden floors.

7.1.1.6 Storage area should always be segregated from packing area.

7.1.1.7 Storage area should be protected with approved fire hydrant system and adequate firefighting measures.

7.1.1.8 Storage area should be equipped with flame detectors for early detection of fire.

7.1.1.9 Power supply to the storage area should be switched off after completion of work.

7.1.1.10 No welding work should be carried out in vicinity of storage area.

7.1.1.11 Smoking should be strictly prohibited in process and godown area. Boards displaying 'No Smoking' should be displayed in all areas and strict enforcement of the same should be ensured.

7.1.1.12 Since chlorine is used and stored in the premises for manufacture of SBP, safety code for chlorine (*see* IS 4263) should be strictly followed.

7.1.2 *Storage of SBP Bags*

7.1.2.1 SBP bags should be stored at a gap of around 1m from electric system. The bags should never be stored below electric fittings or fixtures.

7.1.2.2 SBP bags should not be stored on dirt or damp floors.

7.1.2.3 No combustible material like oily rags should be kept near storage area of SBP bags.

7.1.2.4 SBP bags should be stored out of direct sunlight.

7.1.2.5 The electrical fixtures should be flame proof. The wiring should be preferably underground (concealed) The cables should not have any joints in between.

7.1.2.6 The storage platform should be made at an elevated height from the floor to allow airflow at the bottom also.

7.1.2.7 Proper dry air-cooled storage area should be maintained.

7.1.2.8 Regular temperature monitoring of stacked SBP bags should be done in cooled storage area.

7.1.3 *Storage of SBP Drums*

7.1.3.1 SBP drums should be stored in cool, dry place and away from direct sunlight.

7.1.3.2 The drums should be stored away from source of heat.

7.1.3.3 The drums should be kept closed all the time.

7.1.3.4 The drums should not be stored on dirt or damp floors.

7.1.4 *Bulk Storage of Bleaching Powder in Godowns*

7.1.4.1 The flooring of the godown should be impervious with proper drainage.

7.1.4.2 The bags can be stacked about 10 m high on the floor.

7.1.4.3 The material should be stored in 200 bags to 250 bags (25 kgs weight) lot in line with the batch size.

7.1.4.4 There should be a gap of 15 inches or so between two lots to facilitate movement of personnel.

7.1.4.5 Bags must be stored 8 inch to 10 inch away from the walls to facilitate air circulation.

7.1.4.6 A good number of exhaust fans should be installed in the godown for better ventilation. Only minimum number of them should be run during winter.

7.1.4.7 Humidity in air during monsoon does not affect packed bleaching powder due to double packing; i.e. HDPE laminated bag with liner.

7.1.4.8 During periods of lean sales it is desire able to curtail production rather than overloading the godown.

7.1.4.9 Material should be dispatched from the godown in the order it is received; i.e. material entering first should go out first so that duration of its stay in the godown is optimum.

7.1.4.10 The godown should have two additional doors (facing each other), which should have frames made of M.S. flats / small diameter pipes & M.S, angels to facilitate cross ventilation.

7.1.4.11 Temperature of the material in the bags should be measured with a temperature gun or temperature gauge daily during summer and 2 times to 3 times a week in winter. Any abnormal rise in temperature in a bag indicates an abnormality. It should be removed from the godown.

7.2 Handling

7.2.1 *General*

7.2.1.1 SBP should be handled wearing an approved respirator, chemical resistant gloves, safety goggles, aprons and other protective clothing (see **6**).

7.2.1.2 Sources of ignition such as sparks, smoking and open flames are prohibited where bleaching powder is handled.

7.2.1.3 Persons handling SBP shall strictly follow safety recommendations and safe operating procedures.

7.2.2 *Handling of SBP Bags*

7.2.2.1 Hooks should not be used or dragging not to be followed, while moving SBP bags.

7.2.2.2 The bags should have double liner and tied separately. The outer HDPE bag to be stitched in double rows. Spillage of SBP on bags to be cleaned.

7.2.3 *Handling of SBP Drums*

7.2.3.1 Drums of SBP should not be subjected to rough handling or to abnormal mechanical shock such as dropping, bumping, dragging or sliding.

7.2.3.2 Lifting magnet or sling (rope or chain) which may produce sparks should not be used.

7.2.3.3 Protect drums from any object that will result in abrasion in the surface of the metal.

7.2.3.4 Spillage of SBP on drums should be cleaned by rubbing with a dry cloth urgently.

7.3 Labelling

7.3.1 Any vessel containing stable bleaching powder should carry an identification label or stencil.

7.3.2 Each bag or drum must be labelled with Hazard Class **5.1**.

7.4 Transportation

7.4.1 *Labelling*

Every vehicle used for transporting stable bleaching powder shall be legibly and conspicuously marked with an emergency information panel as per the requirement of *Central Motor Vehicles Act*, 1989 and its amendments thereto.

7.4.2 *Pre-departure*

7.4.2.1 Prior to and after loading, the driver must take a complete inspection of the vehicle to ensure that it meets all performance safety requirements and is road worthy. A few of the numerous items that are checked are the lights, tyres, suspensions and brake.

7.4.2.2 Ensure availability of emergency kit on the vehicle.

7.4.2.3 Ensure availability of transport emergency card (TREM card) and instruction manual during transportation in the vehicle.

7.4.2.4 The availability of GPRS system with SBP loaded vehicles will further improve transport safety as the vehicle movement can be traced.

7.4.2.5 Ensure that the driver carrying SBP is well conversed about the product and its properties

7.4.2.6 SBP should be kept away from heat and moisture.

7.4.2.7 No inflammable material like oil, paint, varnish, grease, etc, should be clubbed with SBP, which may react with SBP and catch fire.

7.4.2.8 Loaded SBP vehicle should be well covered with tarpaulin.

7.4.2.9 No part of SBP bag or drum should come outside from the base of the truck, which can rub with the tyre/any moving part and catch fire.

7.4.2.10 Do not use hooks for loading or unloading.

7.4.2.11 All sources of ignition must be kept away from the truck during transportation.

7.4.2.12 No welding work should be done on SBP loaded Trucks

7.5.3 Driver

Driver should follow specified route only, maintain speed limit, never park the truck near residential areas, drive truck carefully and observe all routes and signals. Avoid overtaking of moving vehicles and do not leave truck without watch at any time. Do not smoke in the SBP loaded truck.

7.5.3.1 In case of emergency (for example, fire, accidents, etc):

a) Stop the engine;

b) Notify police and fire brigade immediately;

c) Mark roads, warn other road users;

d) Keep public away;

e) In case of fire, move the truck immediately to an open area and flush with excess water; and

f) Immediately contact manufacturer/supplier.

7.5.3.2 *Drivers training*

7.5.3.2.1 The basics of any effective safety programme for transportation of SBP is an adequate training.

7.5.3.2.2 The driver should have compulsorily undergone the training provided by RTO for transportation of Hazardous goods

7.5.3.2.3 Each driver must receive adequate amount of classroom training, audio-visual instructions and job training.

7.5.3.2.4 The driver must carry material safety data sheet and TREM card with each consignment of SBP.

7.5.3.2.5 Systematic training/re-training/reviews must be carried out to ensure:

a) Safe driving methods;

b) Actions to be taken during emergency;

c) Proper use of fire extinguishers, kits, TREM cards, instruction manual; and

d) Communication with manufacturer/supplier.

8 SPILLAGE, LEAKAGE AND WASTE DISPOSAL

8.1 General

8.2 Spillage

8.2.1 *General Information*

8.2.1.1 In case of spillage, sweep up spilled substance, but avoid making dust. Remove to safe place.

8.2.1.2 Do not repack the spilled material.

8.2.1.3 Do not bring the spilled material in contact with acid, saw dust or other combustible material.

8.2.1.4 Clean up personnel should wear full protective clothing including breathing apparatus in confined areas, while working in SBP environment.

8.2.1.5 Do not add small amount of water to material where spill has occurred in a confined space or an unventilated building/enclosure and the material is damp and evolving chlorine. The rate of chlorine evolution can be reduced by covering the thinly spread solid with soda ash.

8.2.2 *Land spill (spill on land)*

Avoid discharge of water near the spillage area. Keep combustibles (wood, paper, oil, etc.) away from spilled material. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Recover spilled sodium bleaching powder as much as possible and cover the spilled area with sand.

8.2.2.3 *Containment*

As an immediate precautionary measure, isolate spill or leak area in all directions for at least 50 m (150 ft) for liquids and at least 25 m for solids.

8.2.2.3.1 *Large spill*

Evacuate for at least 100 m.

8.2.2.3.2 *Fire*

In case of fire hazard of the transported vehicle should be isolated in 800 m in all direction.

8.2.2.4 *Consequence*

The release of chlorine gas from the spilled bleaching powder and can affect human beings, animals and environment of the surrounding area. Chlorine gas may travel further in the down wind direction if large quantity of bleaching powder leaks.

8.2.2.5 *Mitigation*

8.2.2.5.1 *Small fire*

Water can be used as extinguisher. There is a limited control with CO_2 or Halon and do not use dry chemicals or foams.

8.2.2.5.2 *Large fire*

Flush or flood it with water from a distance. Avoid moving cargo or vehicle with cargo that is exposed to heat. Cool the containers with water and try to move the storage containers after subsiding the fire area. Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Stay away from the engulfed fire and keep maximum distance while firefighting.

8.2.3 *Water Spill (Spill in Water)*

8.2.3.1 *Containment*

Spilled bleaching powder will mix with water and chlorine will be liberated. chlorine can be neutralised with sodium sulphite, sodium bisulphite, sodium thiosulfate, hydrogen peroxide, and sulphur dioxide.

8.2.3.2 *Mitigation*

In case of spillage of large quantity of SBP into water body, the water of the affected water body may be unfit for consumption and may affect the aquatic life because of high chlorine content. Stop use of contaminated water. Notify proper authorities to stop water intake or to monitor water for contamination

8.2.3.3 *Work Upwind*

8.3 Leakages

8.3.1 *General Information*

In case of a spill or leak, stop the leak as soon as possible.

8.3.2 *Leak from the Truck*

The leaking tanker should be emptied out to another empty tanker or to a storage tank.

8.3.3 *Caution*

Persons handling leaking bleaching powders Tankers should wear proper personal protective equipment.

8.4 Waste Disposal

In the event of decomposition or contamination, the container should not be re-sealed but should be isolated, if possible and flooded with sufficient water to completely dissolve and wash away the contents.

9 FIRE PREVENTION AND FIRE FIGHTING

9.1 General

SBP is not combustible, but it is a strong oxidizer and its heat of reaction with reducing agents may cause ignition. It is thermally unstable, at higher temperatures may undergo accelerated decomposition with release of heat and oxygen. It may readily ignite combustible material. Decomposition can be rapid and violent on heating and contact with incompatible materials.

9.2 Fire Prevention

9.2.1 Eliminate as far as possible all sources of ignition.

9.2.2 Do not store SBP with incompatible materials.

9.2.3 Do not bring in contact with moisture, saw dust and combustible materials, organic matter, etc.

9.2.4 Adequate means of early detection of fire and giving early warnings, alarms shall be provided.

9.2.5 Emergency Response Procedures and Systems

9.2.5.1 Full on-site emergency control procedure and systems shall be established and periodic mock drill shall be carried out.

9.2.5.2 The emergency response team must be well trained with Do's and Don'ts, Standard Operating Procedures (SOP), mitigation systems and hazards of SBP.

9.3 Fire Fighting

9.3.1 In case of fire, keep containers cool by spraying with water, if exposed to fire.

9.3.2 Extinguish with copious amount of water.

9.3.3 Do not use foam, alcohol foam, carbon tetrachloride or dry ammonium compound fire extinguishers or carbon dioxide.

9.3.4 Fight fire from protected location or maximum possible distance.

9.3.5 In the event of fire, wear full protective clothing and self-contained breathing apparatus with full face piece operated in positive pressure mode.

10 TRAINING

10.1 Employee Education and Training

All personal engaged in the operation, maintenance and attending emergency of SBP and related system shall receive suitable training for the work on which they are engaged.

10.1.1 *Training*

10.1.1.1 Training shall cover all aspects and potential hazards that the operator is likely to encounter. Following areas should be covered in the training:

a) Potential health hazards of SBP, Chlorine and Lime;

b) MSDS of the above chemicals;

c) Site safety regulations;

d) Emergency response procedures;

e) Use of protective clothing/breathing apparatus;

f) Accidental release measures;

g) First aid;

h) Fire-fighting;

j) Resource operations; and

k) Communication.

10.1.1.2 In addition, individuals shall receive specific training in the activities for which they are employed. Refresher courses should be arranged on a periodic basis.

10.1.1.3 Each employee should know the location and correct operation of fire alarm systems, sprinklers, other fire-fighting equipment, first aid boxes, etc.

11 HEALTH MANAGEMENT FIRST AID AND MEDICAL TREATMENT

11.1 Health Management

11.1.1 *Personal Hygiene*

Eating, drinking and storing of food near the place, where SBP is handled should be prohibited.

11.1.2 *Physical Examination*

11.1.2.1 *Pre-placement examination*

Proper pre-placement medical examination of a personnel should be carried out to determine the physical fitness before assigning the job.

11.1.2.2 *Periodic examination*

Annual medical examination should be carried out as required by the regulatory bodies, for example, Factories Act/State Factories Rules, etc.

11.2 First Aid Treatment

11.2.1 *General Principles*

First aid treatment should be started at once in all cases of injury. Refer all injured persons to a physician even when the injury appears to be slight. Give the physician a detailed account of the accident.

11.2.2 *Inhalation*

11.2.2.1 Remove the victim from exposure to fresh air.

11.2.2.2 Remove contaminated clothing and loosen remaining clothing.

11.2.1.3 Allow patient to assume most comfortable position and keep warm.

11.2.2.4 If breathing is difficult, give oxygen through qualified person.

11.2.2.5 If not breathing, give artificial respiration. Get medical attention immediately.

11.2.3 *Ingestion*

If swallowing, immediately rinse mouth with water, give water to drink. Do not induce vomiting, never give anything by mouth to an unconscious person, seek immediate medical assistance.

11.2.4 *Eye Contact*

Immediately irrigate with copious quantities of water for at least 15 min. Keep the eyelids open. Get medical attention immediately.

11.2.5 *Skin Contact*

11.2.5.1 Immediately remove contaminated clothing and shoes.

11.2.5.2 Wash contaminated skin with plenty of water for at least 15 min.

11.2.5.3 If swelling, redness, blistering or irritation occurs, seek medical advice.

11.2.6 *Antidote/Dosage*

Seek medical attention immediately.

11.3 Medical Treatment

11.3.1 Treat skin burns by cleaning the affected area with soap and water.

11.3.2 If bleach gets into the eyes, the person should first flush the eyes with water and then seek medical attention

12 ADDITIONAL INFORMATION

12.1 Stable bleaching powder is classified as a Class 8 Dangerous Good. When handling concentrated stable bleaching powder, appropriate precautions should be taken.

12.2 Very serious allergic reaction to SBP is unlikely, but if it occurs seek immediate medical attention. Symptoms of a serious allergic reaction may include rash, itching/swelling (especially of the face/tongue/throat), severe dizziness, trouble in breathing.

12.3 Unauthorised and untrained employees should not be permitted in areas where stable bleaching powder is being handled.

ANNEX A

(*Clause* 2)

IS No.	*Title*
IS 1065	Stable bleaching powder — Specification
(Part 1) : 2019	Part 1 Household and industrial use (*third revision*)
(Part 2) : 2019	Part 2 Treatment of water intended for drinking
IS 2925 : 1984	Specification for industrial safety helmets (*second revision*)
IS 4155 : 2023	Glossary of terms relating to chemical and radiation hazards and hazardous chemicals (*first revision*)
IS 4263 : 2026	Chlorine — Code of Safety (*first revision*)
IS 8519 : 2024	Guide for selection of occupational protective clothing — Body protection (selection, care, and maintenance) (*first revision*)
IS 8520 : 2023/ ISO 19734 : 2021	Eye and face protection — Guidance on selection, use, and maintenance (*first revision*)
IS 8807 : 2025	Selection of industrial safety equipment for protection of arms and hands — Guidelines (*first revision*)
IS 10667 : 2025	Selection of industrial safety equipment for protection of foot and leg — Guide (*first revision*)